CLASS 05

Modern Causal Inference & Linear Regression

GOVT 10: Quantitative Political Analysis

> This chapter bridges causal reasoning and estimation. We start with the modern toolkit of causal inference — potential outcomes, randomized experiments, and difference-in-differences — and then introduce linear regression, the workhorse method for describing how one variable moves with another.

1 Modern Causal Inference

1.1 The Fundamental Problem of Causal Inference

Start with a single voter. Sarah lives in a competitive congressional district. Three days before the election, she receives a campaign mailer from a candidate. On Election Day, she votes. Did the mailer cause her vote?

To answer this question for Sarah, we would need to compare two states of the world: the one we observed (Sarah received the mailer and voted) and the one we did not (Sarah did not receive the mailer; we want to know whether she still would have voted). The trouble is that the second world does not exist. Sarah either received the mailer or she did not. We cannot rewind the election and run it again under different conditions. The counterfactual outcome is permanently unobservable.

This is the **fundamental problem of causal inference**. For every unit, we observe one outcome and lose access to the other. The individual treatment effect is defined as the difference between two potential outcomes, but only one of them is ever realized.

> **Potential Outcomes Notation**
>
> For each unit i we define two potential outcomes:
>
> - $Y_i(1)$ = the outcome we would observe if unit i received the treatment.
> - $Y_i(0)$ = the outcome we would observe if unit i did not receive the treatment.
>
> The individual treatment effect is
>
> $$\tau_i = Y_i(1) - Y_i(0).$$
>
> For any given unit we observe either $Y_i(1)$ or $Y_i(0)$, but never both. One of the two terms in τ_i is always a counterfactual.

```r
counterfactual_table <- tibble(
  voter = c("Sarah", "James", "Priya", "Andre"),
  received_mailer = c(TRUE, TRUE, FALSE, FALSE),
  voted = c(1, 0, 0, 1),
  counterfactual = c("?", "?", "?", "?")
)
counterfactual_table
```

```
# A tibble: 4 x 4
  voter received_mailer voted counterfactual
  <chr> <lgl>           <dbl> <chr>
1 Sarah TRUE                1 ?
2 James TRUE                0 ?
3 Priya FALSE               0 ?
4 Andre FALSE               1 ?
```

> **What this code does:** We build a small table that makes the fundamental problem concrete. Each voter either received the mailer or did not, and each voted or did not. The question marks are not missing data we forgot to collect; they are the counterfactual outcomes that cannot exist. For Sarah, we know she voted with the mailer, but we cannot observe what she would have done without it. For Andre, we know he voted without the mailer, but we cannot observe what he would have done with one. No amount of survey work fills in those cells.

1.2 Why Counterfactuals Matter for Policy

The counterfactual problem is not a philosophical curiosity. It is the central obstacle every policy evaluation must confront. Imagine a governor who introduces an education reform and observes that test scores rise by eight points over her term. Did the reform cause the gain? Only if scores would have been at least eight points lower in the absence of the reform. If scores were rising five points per year nationwide for unrelated reasons, the reform's effect is closer to three. If scores in similar states without the reform rose by twelve points, the reform may have actually slowed growth.

The same logic applies to campaign decisions. A campaign manager who steers her candidate negative in the final week and wins by three points has no way to know whether positive ads would have produced a five-point win, a two-point win, or a loss. The observed outcome is consistent with many different counterfactuals. Without a credible estimate of the counterfactual, no causal claim is possible.

```r
policy_evaluation <- tibble(
  scenario = c("Observed test score",
               "Counterfactual A: scores would have been flat",
               "Counterfactual B: scores would have risen 12 pts"),
  score = c(78, 70, 82),
  implied_effect = c(NA, 78 - 70, 78 - 82)
)
policy_evaluation
```

```
# A tibble: 3 x 3
  scenario                                         score implied_effect
  <chr>                                            <dbl>          <dbl>
```

```
1 Observed test score                           78            NA
2 Counterfactual A: scores would have been flat       70             8
3 Counterfactual B: scores would have risen 12 pts    82            -4
```

What this code does: We tabulate three scenarios for the same observed score of 78. If scores would have been flat at 70, the reform helped by 8 points. If scores would have risen on their own to 82, the reform hurt by 4 points. The point is not which counterfactual is correct; the point is that the conclusion depends entirely on which counterfactual we believe. Credible causal inference is the work of justifying one counterfactual over the others.

1.3 The Solution: Compare Groups

We can never observe both potential outcomes for the same unit, so we settle for the next best thing. We find a group of units that did receive the treatment and a group of comparable units that did not, and we compare their average outcomes. The hope is that the control group's average outcome stands in for what the treatment group's outcome would have been in the absence of treatment.

Whether that hope is reasonable depends entirely on how the groups were formed. If the treated and untreated units are similar in every way that matters for the outcome, the comparison gives us a credible estimate of the causal effect. If they differ systematically, the comparison delivers a confounded estimate. The whole methodological project of modern causal inference is about constructing comparisons in which the two groups really are comparable.

1.4 The Average Treatment Effect

The quantity researchers most often try to estimate is the **Average Treatment Effect**, or ATE. The ATE is the average of the individual treatment effects across the population:

$$\text{ATE} = E[Y_i(1) - Y_i(0)] = E[Y_i(1)] - E[Y_i(0)]. \tag{1}$$

In words: the ATE is the difference in average outcomes between a world in which everyone is treated and a world in which no one is treated. We cannot directly observe either of those worlds, but if our two groups are comparable, we can estimate the ATE from sample averages:

$$\widehat{\text{ATE}} = \bar{Y}_{\text{treated}} - \bar{Y}_{\text{control}}. \tag{2}$$

```
gotv_summary <- gotv_sim %>%
  group_by(treatment) %>%
  summarise(
    n = n(),
    turnout_rate = mean(voted),
    .groups = "drop"
  )
gotv_summary
```

```
# A tibble: 2 x 3
  treatment     n turnout_rate
      <dbl> <int>        <dbl>
1         0   507        0.414
2         1   493        0.505
```

What this code does: We simulate a GOTV experiment in which 1,000 randomly chosen voters were assigned to receive a mailer (`treatment = 1`) or no contact (`treatment = 0`). The treatment-group turnout rate exceeds the control rate by roughly 9.1 percentage points. Because assignment was random, the gap between groups estimates the ATE: how much the mailer raised turnout on average across the population from which the sample was drawn.

The interpretation of the ATE depends on the type of outcome. For a binary outcome like voting, an ATE of 0.08 means the treatment raised the probability of voting by 8 percentage points on average. For a continuous outcome like grade-four test scores, an ATE of 3.5 means the treatment raised scores by 3.5 points on average. The ATE is always an average; some units may have been moved much more than that, others not at all.

1.5 Randomized Experiments

The cleanest way to ensure the treatment and control groups are comparable is to assign units to them at random. A **randomized experiment** (also called a randomized controlled trial, or RCT) is a study in which the researcher determines who is treated by flipping a coin or running a random-number generator. Random assignment is the gold standard for causal inference because it solves the confounding problem by design rather than by adjustment.

The logic is straightforward. When assignment is random, the coin does not know anything about the units. It does not know their age, education, income, prior turnout history, ideology, neighborhood, or any other characteristic. In expectation, the two groups will look identical on every variable: observed and unobserved, measured and unmeasured. The only systematic difference between them will be the treatment itself. Any difference in outcomes can therefore be attributed to the treatment.

```
rct_data %>%
  group_by(treatment) %>%
  summarise(
    n = n(),
    avg_age = mean(age),
    avg_prior_turnout = mean(prior_turnout),
    avg_income = mean(income),
    avg_interest = mean(political_interest),
    .groups = "drop"
  )
```

```
# A tibble: 2 x 6
  treatment     n avg_age avg_prior_turnout avg_income avg_interest
      <dbl> <int>   <dbl>             <dbl>      <dbl>        <dbl>
1         0   300    42.0              1.99     53708.         4.22
2         1   300    42.2              1.92     54104.         3.94
```

> **What this code does:** We check covariate balance across treatment and control. The two groups have nearly identical average age, prior turnout history, household income, and political interest. We did not arrange this balance; random assignment produced it automatically. The same balancing would hold for variables we did not measure—religiosity, personality traits, the political views of one's friends—which is exactly why randomization is so powerful. It balances everything in expectation, not just the variables we thought to collect.

1.5.1 Green and Gerber on GOTV

The most influential application of randomized experiments in modern American political science is the body of work on get-out-the-vote (GOTV) interventions pioneered by Donald Green and Alan Gerber. In a series of field experiments dating from the late 1990s, they randomly assigned registered voters to receive in-person door-to-door canvassing, phone calls, postcards, or no contact, and they measured actual turnout using public voter files. By randomizing contact, they could isolate the effect of each form of mobilization from the confounders that bedevil observational studies (campaigns tend to contact already-likely voters, and likely voters tend to vote whether or not they are contacted).

The findings reshaped the field: in-person canvassing produced large turnout gains (on the order of seven to ten percentage points per contacted voter), phone calls had small but real effects when delivered by volunteers in a conversational style, and impersonal postcards had essentially no effect. The substantive lesson is that personal contact mobilizes voters; the methodological lesson is that you can credibly estimate the size of that effect only when assignment is random.

1.5.2 Audit Studies and Media Studies

Randomized experiments are not limited to GOTV. **Audit studies** send fictional applicants with otherwise identical resumes to real employers, randomizing only the applicant's name or signaled identity, to detect labor-market discrimination. Bertrand and Mullainathan's 2004 study found that resumes with stereotypically Black names received roughly 50 percent fewer callbacks than otherwise identical resumes with stereotypically White names, a gap that observational comparisons of hiring outcomes cannot cleanly identify because applicants self-select into jobs.

Media studies use similar logic. Researchers randomize which subjects see a particular news story, advertisement, or social-media post, and then measure changes in attitudes, knowledge, or behavior. Because exposure is assigned rather than chosen, the comparison delivers a clean estimate of the effect of the content itself, separating it from the kind of person who would have sought the content out on their own.

> **Why Random Assignment Works**
>
> Random assignment solves the three requirements in one stroke:
>
> 1. **Association** is straightforward to measure once we have two groups and an outcome.

2. **Temporal ordering** is guaranteed by design: the treatment is assigned, then the outcome is measured.

3. **No confounding** is achieved in expectation. Because every potential confounder is distributed independently of treatment status, no Z can systematically explain the difference between groups.

This is why a randomized experiment, when feasible, is usually the strongest possible evidence for a causal claim.

1.5.3 Estimating the ATE with Regression

A randomized experiment makes the ATE simple to compute: it is the difference in group means. Linear regression delivers the same number, plus a standard error.

```
model <- lm(turnout ~ treatment, data = experiment_data)
tidy(model)
```

```
# A tibble: 2 x 5
  term        estimate std.error statistic   p.value
  <chr>          <dbl>     <dbl>     <dbl>     <dbl>
1 (Intercept)    49.6      0.622     79.8  7.37e-286
2 treatment       8.29     0.899      9.21 8.72e- 19
```

What this code does: We regress a simulated turnout score on a treatment indicator. The intercept (about 50) is the control-group mean. The treatment coefficient (about 7) is the ATE: how much higher the treated group's average is than the control group's. The standard error and p-value tell us how confident to be that the gap is not zero. Regression and the simple difference-in-means give exactly the same point estimate; regression adds the uncertainty quantification we will need in later chapters.

1.6 When Experiments Are Impossible

Randomized experiments are powerful, but they are not always available. We cannot randomly assign which states pass a minimum-wage increase, which countries host refugees, which neighborhoods receive new policing strategies, or which voters get exposed to a major news event. Many of the most important questions in political science involve treatments that researchers cannot manipulate. When randomization is off the table, we need other strategies that exploit natural variation in the world to approximate the comparison a randomized experiment would deliver.

1.7 Difference-in-Differences

Difference-in-differences, or DiD, is the most widely used such strategy. It exploits two pieces of structure that often exist in policy data: a treatment that is implemented at a known point in time and applied to some units but not others, and outcome measurements available both before and after that point for both groups. By comparing how each group changed over time, DiD removes confounders that affect both groups equally and isolates the effect of the treatment.

The intuition is best built with a non-political example. Imagine your friend Alex joins a gym in January and loses 15 pounds by June. Did the gym cause the weight loss? Maybe, but during the same months Alex started a stressful job, moved to a new neighborhood, and went through a breakup. Any of these changes could explain some or all of the 15 pounds. Now suppose we have a comparison friend, Jordan, who is similar to Alex in every way and experienced the same life changes but did not join a gym. Jordan lost 8 pounds. The 8 pounds represent what would have happened to Alex without the gym; the additional 7 pounds Alex lost are attributable to the gym itself.

That double subtraction is the heart of DiD. We measure how the treatment group changed over time, measure how the control group changed over time, and subtract the latter from the former. The control group's change captures the common trend; what is left over is the treatment effect.

$$\widehat{\text{DiD}} = (\bar{Y}_{\text{treat, after}} - \bar{Y}_{\text{treat, before}}) - (\bar{Y}_{\text{ctrl, after}} - \bar{Y}_{\text{ctrl, before}}). \tag{3}$$

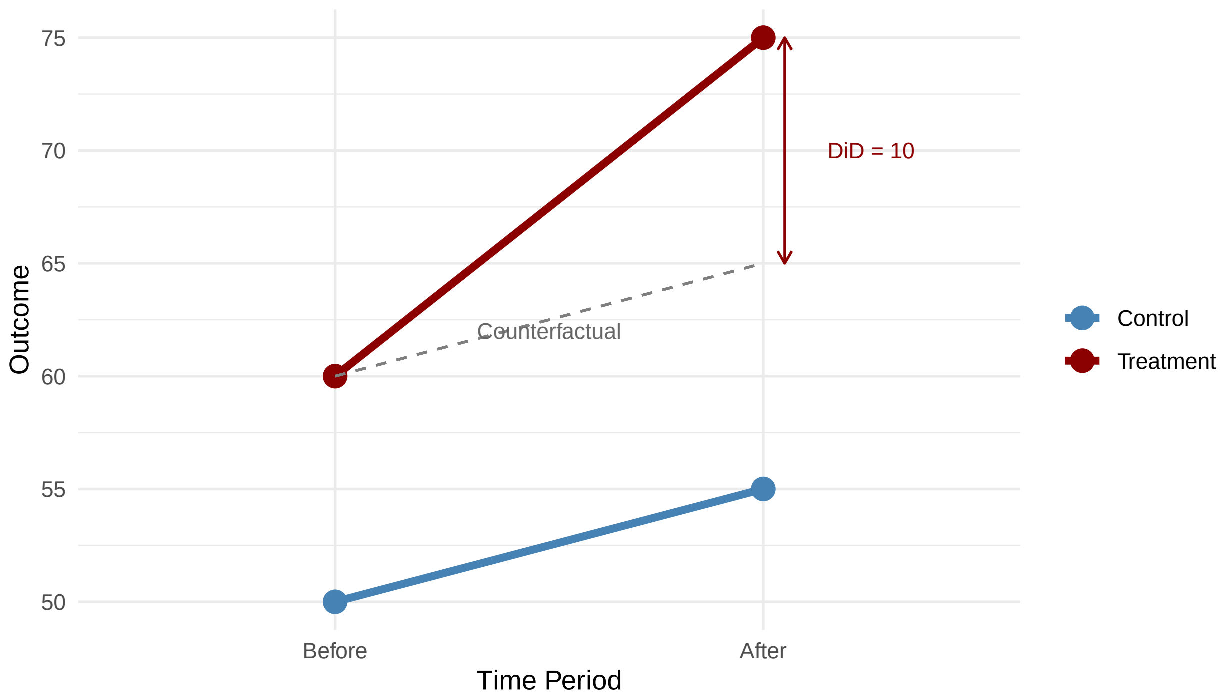

Figure 1: Figure 4.2: The DiD design. The dashed line shows what the treatment group's outcome would have been without treatment, assuming parallel trends. The treatment effect is the gap between the observed outcome and this counterfactual.

In this example, the treatment group rises from 60 to 75 (a change of +15) and the control group rises from 50 to 55 (a change of +5). The DiD estimate is $15 - 5 = 10$. We interpret the control group's 5-point rise as the general trend that would have raised the treatment group anyway, and the additional 10 points as the treatment effect.

```r
did_calculation <- tibble(
  group = c("Treatment", "Treatment", "Control", "Control"),
  period = c("Before", "After", "Before", "After"),
  outcome = c(60, 75, 50, 55)
)
did_calculation
```

```
# A tibble: 4 x 3
  group     period outcome
  <chr>     <chr>    <dbl>
1 Treatment Before      60
2 Treatment After       75
3 Control   Before      50
4 Control   After       55
```

```r
treatment_change <- 75 - 60
control_change <- 55 - 50
did_effect <- treatment_change - control_change
did_effect
```

```
[1] 10
```

> **What this code does:** We carry out the DiD calculation step by step. The treatment group's change (15 points) captures both the treatment effect and the common trend. The control group's change (5 points) captures only the common trend. Subtracting yields the treatment effect: 10 points. This is the same logic as the gym example, applied to numbers rather than weight loss.

1.8 Card and Krueger: Minimum Wage in New Jersey

The DiD design rose to prominence through David Card and Alan Krueger's 1994 study of minimum-wage effects. Standard economic theory predicted that raising the minimum wage would reduce low-wage employment: forced to pay higher wages, employers would hire fewer workers. The empirical evidence to that point was largely correlational and broadly consistent with this prediction, though plagued by confounding.

In 1992, New Jersey raised its minimum wage from $4.25 to $5.05. Neighboring Pennsylvania kept its rate at $4.25. Card and Krueger surveyed fast-food restaurants in both states before and after the increase. They were not the first to study minimum wages; what distinguished their work was the design. Rather than comparing New Jersey to itself before and after (which would confound the policy with state-wide economic trends) or comparing New Jersey to Pennsylvania after the change (which would confound the policy with all the pre-existing differences between the states), they compared the *change* in New Jersey employment to the *change* in Pennsylvania employment.

Their result was striking. Employment in New Jersey fast-food restaurants did not fall after the minimum-wage increase; if anything, it rose slightly relative to Pennsylvania. The DiD estimate ran in the opposite direction from the textbook prediction. The finding launched decades of follow-up research and helped establish DiD as a standard tool in economics and political science.

```r
nj_pa_did
```

```
# A tibble: 4 x 3
  state        period employment
  <chr>        <chr>       <dbl>
1 New Jersey   Before       20.4
2 New Jersey   After        21
3 Pennsylvania Before       23.3
4 Pennsylvania After        21.2
```

```r
nj_change <- nj_pa_did$employment[nj_pa_did$state == "New Jersey" &
                                    nj_pa_did$period == "After"] -
             nj_pa_did$employment[nj_pa_did$state == "New Jersey" &
                                    nj_pa_did$period == "Before"]
pa_change <- nj_pa_did$employment[nj_pa_did$state == "Pennsylvania" &
                                    nj_pa_did$period == "After"] -
             nj_pa_did$employment[nj_pa_did$state == "Pennsylvania" &
                                    nj_pa_did$period == "Before"]

did_nj_pa <- nj_change - pa_change
did_nj_pa
```

```
[1] 2.7
```

> **What this code does:** We replicate the Card–Krueger calculation with stylized employment numbers. Fast-food employment in New Jersey rose slightly after the minimum-wage increase, while employment in Pennsylvania fell. The DiD estimate (New Jersey's change minus Pennsylvania's change) is positive, suggesting the minimum-wage increase did not cost jobs on net. The substantive finding remains debated; the methodological innovation is that the comparison removes any confounder that affected both states equally, including the broader business cycle.

1.9 The Parallel-Trends Assumption

DiD relies on a critical assumption: **in the absence of treatment, the treatment and control groups would have followed parallel trends**. The groups can start at different levels (New Jersey had different fast-food employment levels than Pennsylvania), but they must change at the same rate when no treatment is applied. The control group's change then serves as a stand-in for what would have happened to the treatment group.

This assumption is fundamentally about the counterfactual: we are claiming that the control group's change is what the treatment group's change would have been without the treatment. We can never test this directly, because the counterfactual is unobservable. But we can check whether the groups moved in parallel in periods *before* the treatment was applied. If the trends were diverging before the policy took effect, the parallel-trends assumption is implausible, and the DiD estimate is suspect.

> **When Parallel Trends Fails**
>
> The classic failure case is when the treatment group is on a different trajectory before treat-

> ment begins. If New Jersey's economy was already growing faster than Pennsylvania's before the minimum-wage increase, then comparing changes would credit (or blame) the policy for what was really pre-existing divergence. Always plot the pre-treatment trends before trusting a DiD estimate.

1.10 DiD with Regression

For DiD, the regression specification has three terms: a treatment-group indicator (does this unit belong to the group that eventually got treated?), a post-period indicator (is this observation from after the treatment date?), and the interaction between them. The interaction coefficient is the DiD estimate.

```
did_model <- lm(outcome ~ treat + post + treat:post,
                data = did_regression_data)
tidy(did_model)
```

```
# A tibble: 4 x 5
  term        estimate std.error statistic   p.value
  <chr>          <dbl>     <dbl>     <dbl>     <dbl>
1 (Intercept)    49.2      0.722     68.1  2.31e-138
2 treat           6.14     0.974      6.31 1.81e-  9
3 post            2.76     1.02       2.71 7.38e-  3
4 treat:post      8.49     1.38       6.17 3.89e-  9
```

> **What this code does:** We fit a DiD regression with three terms. `treat` captures the baseline gap between treatment and control groups before the policy. `post` captures the change over time that affected both groups. The interaction `treat:post` captures the *additional* change in the treatment group beyond the common trend—the DiD estimate. In our simulated data this is roughly 8, matching the true effect we built into the simulation. In real applications, the standard error on the interaction tells us whether the estimated effect is distinguishable from zero.

1.11 DiD in Practice: Policy Evaluation Examples

DiD has become one of the most widely used designs in applied policy research. Three examples illustrate the range of questions it can address.

Medicaid expansion. The Affordable Care Act made Medicaid expansion optional for states. Some expanded in 2014; others did not. Researchers compared changes in uninsured rates from before to after 2014 in expansion versus non-expansion states. The DiD estimate isolates the effect of expansion from the broader trends in health-insurance coverage that were affecting all states.

Bail reform. Several jurisdictions (Philadelphia, New York City, Cook County, Harris County) eliminated cash bail for low-level offenses between 2017 and 2019. Comparing pretrial detention rates in reforming jurisdictions to detention rates in similar non-reforming jurisdictions, before and after the policy change, isolates the effect of the reform from nationwide trends in pretrial practice.

Refugee integration. When countries adopt new integration policies (language training, work-permit reforms, settlement assistance), researchers can sometimes compare changes in employment or language acquisition among refugees in the policy-changing country to changes among refugees in similar countries that kept their policies fixed. The design controls for the broader macroeconomic conditions that affect all refugees in the region.

In each case, the design works only if the parallel-trends assumption is credible. Investigators support the assumption with pre-trend plots, with arguments about why the timing of the policy was unrelated to pre-existing trends, and sometimes with placebo tests on outcomes that should not have been affected.

1.12 How the Toolkit Maps Onto the Three Requirements

The methods in Segment B are best understood as ways of satisfying the three requirements from Segment A in increasingly demanding settings.

Requirement	**Randomized experiment**	**Difference-in-differences**
Association	Compare means across groups	Compare changes across groups
Temporal ordering	Treatment assigned, then outcome measured	Pre/post timing built into the design
No confounding	Randomization balances Z in expectation	Common trends differenced out

Table 1: How experiments and DiD address the three requirements.

Randomization is the more powerful tool when feasible because it addresses confounding by design: the assignment mechanism is independent of every potential Z. DiD is the more flexible tool because it can be applied to policy variation that the researcher could never randomize: differences across states, jurisdictions, or countries that arise from the political process itself. The cost is that the credibility of DiD rests on the parallel-trends assumption, which is an assumption rather than a feature of the design.

1.13 Common Mistakes Revisited

The mistakes from Segment A reappear in Segment B in subtler forms. Selection bias becomes the violation of parallel trends. Reverse causation becomes anticipation effects (units adjust before the policy takes effect because they see it coming). Confounding becomes any time-varying factor that affects one group differently from the other. The vocabulary shifts but the underlying logic is the same: a credible causal claim requires a credible counterfactual, and a credible counterfactual requires that nothing other than the treatment differs systematically between the groups being compared.

Mistake 5: Heterogeneous treatment effects ignored

The ATE is an average; it hides variation. A GOTV mailer might have a large effect on infrequent voters and almost no effect on habitual voters. A minimum-wage increase might raise some workers' wages and price others out of the labor market. Always ask whether the treatment effect is likely to differ across subgroups, and report subgroup analyses when they are substantively important.

2 Linear Regression

Why This Matters

Imagine you work for a campaign and need to predict how vote share will respond to advertising. Or you're advising a White House team that wants to know how presidential approval moves with the economy. Regression turns these vague hunches into concrete numbers: "For each percentage point increase in unemployment, presidential approval falls by about 1.5 points." Without regression, you're guessing. With it, you can quantify relationships, make predictions, and reason carefully about uncertainty.

2.1 What Makes a Relationship Linear?

A linear relationship means that as one variable increases by a fixed amount, another variable changes by a constant amount, regardless of where you start. If each additional year of education raises political knowledge by the same number of points whether we are comparing a 9th grader to a 10th grader or a college junior to a senior, the relationship is linear. The rate of change is constant across the entire range of the predictor.

This constant rate of change is what distinguishes linear relationships from curved ones. In a curved relationship, the effect of an additional unit of X depends on where you start. The effect of an extra dollar of campaign spending might be much larger for an unknown challenger than for an entrenched incumbent who is already saturated with name recognition. Linear regression cannot capture such patterns directly, though we can extend it with transformations or interactions when needed.

In practice, many political science relationships are approximately linear over the range of data we observe. Even when the true relationship is slightly curved, a straight line often gives a useful first-pass summary. The question is whether the linear approximation is good enough for the question you are asking.

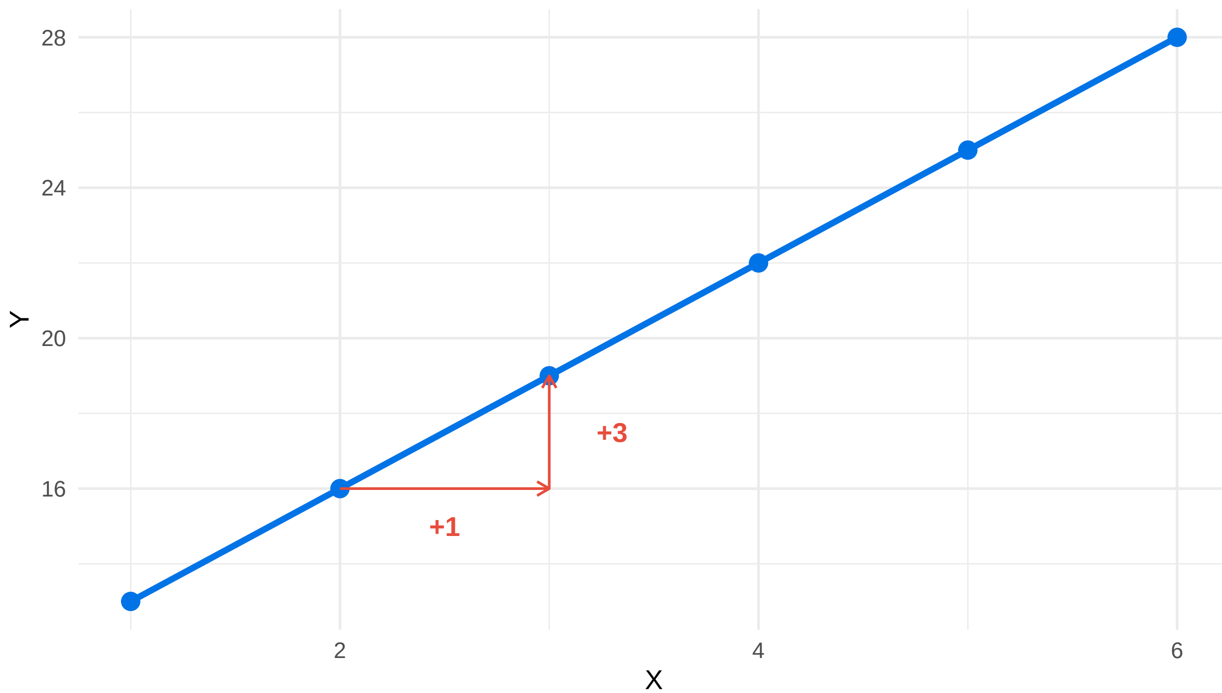

Figure 2: Figure 5.1: A linear relationship has a constant rate of change. Each one-unit increase in X produces the same change in Y.

2.2 The Linear Model

The basic linear model describes the relationship between an outcome Y and a predictor X with a simple equation:

$$Y = \beta_0 + \beta_1 X + \epsilon \tag{4}$$

Here β_0 is the intercept, the predicted value of Y when X equals zero. β_1 is the slope, telling us how much Y changes when X increases by one unit. The term ϵ is the error, capturing everything that affects Y but is not in the model: omitted variables, measurement noise, and the inherent unpredictability of human behavior.

The job of regression is to find the values of β_0 and β_1 that best describe the relationship between X and Y in our data. Different definitions of “best” lead to different estimators. The most common is Ordinary Least Squares (OLS), which picks the line that minimizes the sum of squared vertical distances between observed points and the fitted line.

Symbol	Name	Meaning
Y	Dependent variable	Outcome we explain
X	Independent variable	Predictor
β_0	Intercept	Value of Y when X = 0
β_1	Slope	Change in Y per unit of X
ϵ	Error term	Unexplained variation

Table 2: Components of the Linear Model

2.3 Ordinary Least Squares

The Goal

We want a line through our data that fits "best." But what does "best" mean? The intuition: we want predictions that are as close as possible to the actual data. The question is how to measure "close."

Your first instinct might be to add up all the errors (residuals) and pick the line that makes the sum smallest. The problem is that positive and negative errors cancel. A line that misses every point by ten units — half above, half below — has residuals summing to zero, but it is clearly a terrible line.

You could fix this by using absolute values: $|+5| + |-2| + |+2| + |-5| = 14$. This works, and a method called Least Absolute Deviations does exactly this. But absolute values create a different problem: the function has a "corner" at zero, which makes the math harder, and it treats all errors as proportionally equal — missing by 10 is just twice as bad as missing by 5.

Squaring solves both problems. Squared errors cannot cancel, the function is smooth and easy to differentiate, and large errors get penalized more than small ones (a miss of 10 contributes 100 to the sum, while a miss of 5 contributes only 25). OLS chooses the intercept and slope that minimize the sum of squared residuals:

$$\min_{\beta_0, \beta_1} \sum_{i=1}^{n} (Y_i - \hat{Y}_i)^2 \tag{5}$$

Intuition summary: OLS finds the line that minimizes the sum of squared vertical distances from each point to the line. Large errors hurt much more than small ones, so OLS tries especially hard to avoid big mistakes. When someone says "regression," they almost always mean OLS regression.

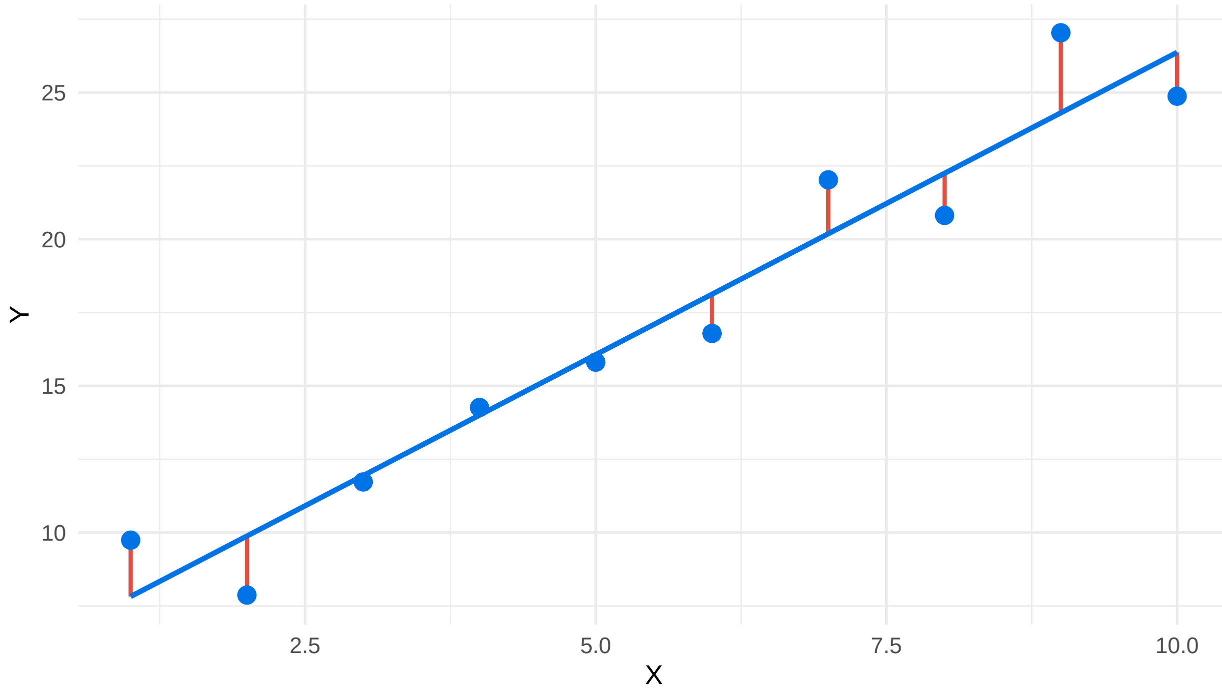

Figure 3: Figure 5.2: OLS minimizes the sum of squared residuals — the vertical distances from points to the line.

2.4 Population Parameters and Sample Estimates

There is an important distinction between the true relationship in the population and our estimate of it from a sample. We use Greek letters for population parameters and add "hats" to denote sample estimates. The true population model uses parameters we never observe directly:

$$Y_i = \alpha + \beta X_i + \mu_i \tag{6}$$

Our estimated model uses statistics calculated from the sample we actually have:

$$Y_i = \hat{\alpha} + \hat{\beta} X_i + \hat{\mu}_i \tag{7}$$

The hat notation reminds us that estimates are approximations. If we collected a different random sample from the same population, we would get slightly different estimates. This sampling variability is quantified by the standard error, which measures how much an estimate would typically vary across repeated samples.

2.5 Interpreting Coefficients

The slope coefficient tells us how many units Y changes when X increases by one unit. Suppose we have data on the relationship between economic performance and presidential approval, where each observation is a quarter of a presidential term.

```
model_approval <- lm(approval ~ unemployment, data = approval_econ)
approval_tidy <- tidy(model_approval)
approval_tidy
```

```
# A tibble: 2 x 5
  term         estimate std.error statistic  p.value
  <chr>           <dbl>     <dbl>     <dbl>    <dbl>
1 (Intercept)     63.9      2.01       31.9  1.74e-46
2 unemployment    -1.89     0.288      -6.57 5.11e- 9
```

> **What this code does:** We regress presidential approval on the unemployment rate. The slope of about -1.89 means that each one-percentage-point increase in unemployment is associated with a 1.89-point drop in approval. The intercept of about 65 is the predicted approval rate when unemployment is zero — a value outside any plausible range, so we focus our interpretation on the slope.

The units of the slope depend on the units of both variables. If we measured unemployment as a fraction (0.05 rather than 5%), the slope would be 100 times larger but the substantive story would be identical. Rescaling changes the number, not the meaning.

The intercept is the predicted value of Y when X equals zero. In many political examples this is not substantively meaningful — zero unemployment, zero campaign spending, zero years of education — so we report the intercept for completeness but focus our interpretation on the slope.

When the predictor is binary, taking values 0 or 1, the slope has a particularly clean interpretation: it is simply the difference in means between the two groups.

```
turnout_model <- lm(voted ~ college, data = turnout_data)
tidy(turnout_model)
```

```
# A tibble: 2 x 5
  term        estimate std.error statistic   p.value
  <chr>          <dbl>     <dbl>     <dbl>     <dbl>
1 (Intercept)    0.469   0.00757      61.9 3.08e-172
2 college        0.185   0.0109       16.9 1.44e- 45
```

> **What this code does:** We regress turnout (the probability someone voted) on a binary indicator for college education. The intercept is the average turnout among non-college respondents. The slope is the gap: college graduates are about 20 percentage points more likely to vote than non-graduates. With a binary predictor, OLS just compares group means.

2.6 Standard Errors and t-Statistics

A coefficient by itself is only half the story. We also need to know how precisely we have estimated it. The standard error measures how much our estimate would typically vary if we drew a different sample. The t-statistic is the coefficient divided by its standard error: it tells us how many standard errors our estimate sits away from zero.

A common rule of thumb is that t-statistics with absolute values above roughly 2 correspond to relationships that are statistically distinguishable from zero at the 5% level. The p-value reported alongside the t-statistic translates this into a probability: the probability of seeing a coefficient at least this large if the true effect were exactly zero. Smaller p-values mean stronger evidence against the null of "no relationship."

> **Reading regression output step-by-step:**
>
> **Step 1:** Find the row you care about. Each row is a coefficient. The first row is always `(Intercept)`. The next rows are your predictors.
>
> **Step 2:** Read `estimate`. This is the slope or intercept. For the slope: "for each 1-unit increase in X, Y changes by [estimate] units."
>
> **Step 3:** Check `std.error`. Smaller means more precise. Across repeated samples, the estimate would typically vary by about $\pm$ this much.
>
> **Step 4:** Look at `statistic`. This is the t-statistic = estimate / std.error. Absolute values above about 2 suggest the coefficient is reliably different from zero.
>
> **Step 5:** Check `p.value`. If $p < 0.05$, the coefficient is statistically significant at the 5% level.
>
> **Template sentence:** "Each one-point increase in unemployment is associated with a 1.89-point decrease in presidential approval ($p < 0.05$)."

2.7 R-Squared: Variance Explained

R-squared measures the proportion of variation in Y that is explained by X. It ranges from 0 (X explains nothing) to 1 (X explains everything).

$$R^2 = 1 - \frac{\sum(Y_i - \hat{Y}_i)^2}{\sum(Y_i - \bar{Y})^2} \tag{8}$$

The numerator is the residual sum of squares — the variation left over after fitting the line. The denominator is the total sum of squares — the variation in Y before considering X at all. The ratio is the fraction of variation that remains unexplained, and subtracting from 1 gives us the fraction explained.

What counts as a "good" R-squared depends entirely on context. Physics experiments routinely produce R-squared values above 0.95 because conditions are controlled and confounders are few. Survey-based political research often produces R-squared values between 0.10 and 0.30, because human attitudes and behaviors are driven by dozens of factors no single model captures.

Context	Typical R^2	Why
Physics experiments	0.95–0.99	Controlled conditions, few confounders
Lab psychology	0.40–0.70	Some control, but human variation remains
Survey research	0.10–0.30	Many factors influence opinions/behavior
Election prediction	0.60–0.85	Fundamentals matter, but shocks intrude
Stock markets	0.01–0.05	Near-random; predictability gets arbitraged away

Table 3: Benchmarks for R-squared by field. Compare your R-squared to similar studies, not to physics.

> **How to evaluate your R-squared:**
>
> **Do NOT say:** "My R-squared is only 0.15, so my model is bad."
>
> **DO say:** "My R-squared of 0.15 means education explains about 15% of the variation in turnout. The other 85% comes from factors not in my model: age, race, mobilization, weather, district competitiveness. Compared to other survey-based studies, this is a typical and meaningful result."
>
> A statistically significant slope with a modest R-squared still tells you something real: X is reliably associated with Y, even if X is far from the only thing that matters.

2.8 Running Regressions in R

The `lm()` function fits linear models in R. The basic syntax specifies the outcome, a tilde, the predictor, and a data frame:

```
model <- lm(approval ~ unemployment, data = approval_econ)
model
```

```

Call:
lm(formula = approval ~ unemployment, data = approval_econ)

Coefficients:
 (Intercept)  unemployment
      63.921        -1.892
```

> **What this code does:** We fit a linear model predicting approval from unemployment. Printing the model object shows the estimated intercept and slope. That's useful for a quick look, but for serious work we want more detail.

The `tidy()` function from the broom package produces clean, tidy output including coefficient estimates, standard errors, t-statistics, and p-values:

```
tidy(model)
```

```
# A tibble: 2 x 5
  term         estimate std.error statistic  p.value
```

```
  <chr>            <dbl>     <dbl>     <dbl>    <dbl>
1 (Intercept)      63.9      2.01      31.9  1.74e-46
2 unemployment     -1.89     0.288     -6.57 5.11e- 9
```

The `glance()` function from the same package gives model-level statistics including R-squared, adjusted R-squared, and the residual standard deviation:

```r
glance(model)
```

```
# A tibble: 1 x 12
  r.squared adj.r.squared sigma statistic       p.value    df logLik   AIC   BIC deviance
      <dbl>         <dbl> <dbl>     <dbl>         <dbl> <dbl>  <dbl> <dbl> <dbl>    <dbl>
1     0.356         0.348  4.70      43.2 0.00000000511     1  -236.  479.  486.    1721.
# i 2 more variables: df.residual <int>, nobs <int>
```

> **Template for reporting your model:** ”The model explains [r.squared × 100]% of the variation in approval. A one-point increase in unemployment is associated with a [estimate] point change in approval (SE = [std.error], p = [p.value], n = [nobs]).”

3 Using AI for This Material

Where AI Helps This Week

- Explaining the difference between association and causation in a worked example, including spelling out the counterfactual a study is implicitly comparing against.
- Defining DAGs, confounders, mediators, and colliders in the abstract and walking through textbook examples of each.
- Articulating what the SUTVA (no-interference) assumption requires and where it tends to fail.

Where AI Gets This Wrong This Week

- Identifying confounders for a specific empirical setting requires substantive knowledge of the political context, which AI does not have.
- Evaluating whether the parallel-trends assumption holds in your data requires looking at pre-treatment trends, which AI cannot inspect.
- It often fails to recognize post-treatment bias when you describe a ”control” variable that is really an outcome of treatment.

A prompt that works for this week:

> I am studying whether televised debates change voter preferences. List the most likely confounders that would make a simple correlation between debate-watching and vote

choice misleading, and explain why each one matters.

A prompt that misleads:

> Here is my dataset on minimum-wage increases and county employment. Did the policy cause the change?

Why it misleads: causal identification requires evaluating an assumption (parallel trends, ignorability, exclusion) against the specific data-generating process, which AI cannot inspect — it will return a generic difference-in-differences answer without asking whether the design is credible.

> **Where AI Helps This Week**
>
> - Writing correct `lm()` syntax for a stated specification, including interactions and factor variables.
> - Explaining the sign and rough magnitude of a coefficient given the units of the variables.
> - Translating a fitted regression equation into a one-sentence interpretation suitable for a paper.

> **Where AI Gets This Wrong This Week**
>
> - Deciding which controls to include is a theory question: adding "all available controls" is wrong, omitting confounders is wrong, and including post-treatment variables is wrong — AI cannot tell which is which in your model.
> - It cannot diagnose Simpson's paradox in your data without seeing the subgroup means.
> - It cannot tell you whether a continuous coefficient is "large" without knowing the variable's units and the substantive range you care about.

A prompt that works for this week:

> Explain in two sentences what the coefficient on `spending_thousands` means in the model `lm(vote_share ~ spending_thousands + incumbent, data = races)`.

A prompt that misleads:

> Which control variables should I include in my regression of turnout on campaign contact?

Why it misleads: control selection is a theory and design question — what is a confounder versus a mediator versus a collider in *your* causal story — and AI will produce a confident kitchen-sink list that mixes valid controls with bias-inducing ones.

4 Common Mistakes and How to Avoid Them

Mistake 1: Interpreting Regression as Causation

A statistically significant regression coefficient shows that X and Y are associated, possibly after controlling for some other variables. It does not prove that X causes Y. Causation requires a research design that handles unobserved confounders. Always distinguish association from causation, especially in observational data.

Mistake 2: Extrapolating Beyond the Data

The regression line describes the relationship within the range of X you observed. Predictions for X values far outside this range are unreliable — the linear approximation may not hold. If unemployment in your sample ranges from 3.5% to 9.5%, do not use the model to predict approval at 20% unemployment.

Mistake 3: Controlling for Mediators or Post-Treatment Variables

Adding ”more controls” is not always better. Variables that are consequences of X — or that were measured after X happened — can introduce bias rather than remove it. Think carefully about whether a candidate control variable is a common cause or a downstream effect.

Mistake 4: Dismissing Low R-Squared

In observational political research, R-squared values of 0.10 to 0.30 are common and can still reveal important relationships. A statistically significant slope with modest R-squared tells you that X is reliably associated with Y, even if X is far from the only thing that matters.

5 R Functions Reference

Function	Purpose	Example
`cor()`	Correlation between two variables	`cor(x, y)`
`lm()`	Fit a linear model	`lm(y ~ treat, data)`
`tidy()`	Tidy regression output	`tidy(model)`
`group_by()`	Group rows for summarization	`group_by(treatment)`
`summarise()`	Collapse groups to summaries	`summarise(mean(y))`
`pivot_wider()`	Reshape long data to wide	`pivot_wider(...)`

Table 4: Key R functions used in Chapter 4.

Function	Purpose	Example
`lm()`	Fit linear model	`lm(y ~ x, data)`
`lm()` (multi)	Add controls	`lm(y ~ x + z, data)`
`tidy()`	Clean coefficient table	`tidy(model)`
`glance()`	Model-level statistics	`glance(model)`
`augment()`	Add fitted values & residuals	`augment(model)`
`predict()`	Make predictions	`predict(model, newdata)`
`coef()`	Extract coefficients	`coef(model)`
`confint()`	Confidence intervals	`confint(model)`

Table 5: Key R Functions for Linear and Multivariate Regression

6 Practice Problems

Question 1: Confounder Hunt

A study reports that congressional districts represented by women have lower defense spending requests. The author concludes that female representatives cause lower defense spending. Identify at least two plausible confounders that could generate this association without a causal effect of representative gender. For each, briefly describe the mechanism.

Question 2: Designing a GOTV Experiment

A city wants to know whether a new text-message GOTV program increases turnout. Sketch a randomized experiment to evaluate the program. Specify the population, the unit of randomization, the treatment, the control condition, and the outcome. What is the ATE you would estimate, and how would you compute it?

Question 3: A DiD Calculation

A neighboring-states study of minimum wage finds that fast-food employment in the treatment state rose from 21.0 to 22.3, while employment in the comparison state fell from 22.5 to 21.1. Compute the DiD estimate and explain what it implies about the effect of the minimum-wage increase. What assumption must hold for this estimate to be credible?

Question 4 (Computational)

Simulate a randomized experiment with 800 units. Randomly assign half to treatment. Generate an outcome equal to 50 plus 4 times the treatment indicator plus normally distributed noise with standard deviation 8. Compute the ATE by (a) taking the difference in group means and (b) running a linear regression of outcome on treatment. Verify that the two approaches give the same point estimate, and report the regression's standard error on the treatment coefficient.

Question 5: Critique a Media-Study Claim

A widely shared op-ed argues that ”people who watch cable news are more politically polarized, so cable news is polarizing the country.” Apply the three requirements from Segment A and the potential-outcomes framework from Segment B to critique this claim. What design would you propose to estimate the causal effect of cable-news exposure on polarization?

For Further Study

Students interested in going deeper should look at Joshua Angrist and Jorn-Steffen Pischke's *Mastering 'Metrics*, which offers an accessible treatment of randomized experiments and difference-in-differences with detailed empirical examples. Judea Pearl's *The Book of Why* provides a complementary philosophical perspective on causation and the limits of correlation-based reasoning. For political-science applications specifically, Don Green and Alan Gerber's *Get Out the Vote* summarizes a remarkable program of GOTV field experiments and demonstrates what credible causal inference looks like in practice.

Question 1

A simple regression of presidential approval on the unemployment rate yields a slope of -1.8. Interpret this number in plain English. Why might this estimate be biased relative to the true effect of unemployment on approval?

Question 2

A campaign analyst runs a regression of vote share on advertising spending and finds a slope of 0.12 (12 votes per $1,000). When she adds incumbency status to the model, the slope drops to 0.05. What does this change tell you? Which estimate is more credible, and why?

Question 3

A colleague proposes that, to estimate the effect of education on political knowledge, the model should also include income (since income is correlated with both). Another colleague argues that income is a consequence of education, so controlling for it would be a mistake. Who is right? Why does the answer matter for the size of the estimated education coefficient?

Question 4 (Computational)

Simulate 200 observations where $Y = 5 + 2X + 3Z + \epsilon$ with X and Z positively correlated. Fit both the simple regression of Y on X and the multivariate regression of Y on X and Z. Compare the X coefficients. Which one is closer to the true value of 2? Why?

For Further Study

This chapter introduced simple and multivariate linear regression. Next week we extend the framework to categorical predictors and interaction terms, which let the effect of one variable depend on the value of another. For deeper treatment, Wooldridge's *Introductory Econometrics* gives a rigorous account of omitted variable bias; Kellstedt and Whitten's *The Fundamentals of Political Science Research* offers an intuitive walkthrough with political examples; and Angrist and Pischke's *Mostly Harmless Econometrics* connects regression to the broader project of causal inference from observational data.